Simplification of Real Elementary Functions

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ABSTRACT

We describe an algorithm, based on Risch's real structure theorem, that determines explicitely all the algebraic relations among a given set of real elementary functions. We also provide examples from its implementation that illustrate the advantages over the use of complex logarithms and exponentials.

INTRODUCTION

An *elementary function* of a variable x is a function that can be obtained from the univariate rational functions in x, by repeatedly adjoining a finite number of nested logarithms, exponentials, and algebraic functions. Since the mid 19th century, mathematicians have been trying to determine whether a given set of elementary functions is algebraically independent (see for example (Liouville 1837) where the transcendence and independence of $log(x)$ and e^x is proven). For example[1], $\mathbf{Q}(x, \log(x), e^{\log(x)/2})$ has transcendence degree 2 and not 3 over $\mathbf{Q}$ since $e^{\log(x)/2}$ satisfies $y^2 - x = 0$. Risch (1979) gave a theorem and an algorithm that found explicitly all the algebraic relations among a set of logarithms and exponentials over $K(x)$ where $K \subset \mathbf{C}$ is a finitely generated extension of $\mathbf{Q}$ (provided that the algebraic structure of K over $\mathbf{Q}$ and of its constant extensions generated by the algorithm remains known). Applying this algorithm requires the trigonometric functions to be converted to complex exponentials and logarithms, which often yields complex algebraic relations among real elementary functions. For example, in order to compute

$$\int \frac{\tan(\frac{\arctan(x)}{3})^2 + 1}{x^2 + 1} dx$$

one first converts the integrand to

$$\frac{4\alpha}{(x^2+1)(\alpha+1)^2}$$

where

$$\alpha = e^{-\log(\frac{\sqrt{-1}+x}{\sqrt{-1}-x})/3}.$$

Applying Risch's algorithm to $\mathbf{Q}(x, \alpha)$ yields

$$\alpha = \left(\frac{\sqrt{-1}+x}{\sqrt{-1}-x}\right)^{-1/3}$$

so the integrator returns a complex algebraic function, while, in fact, $y = \tan(\frac{\arctan(x)}{3})$ is a real algebraic function satisfying $y^3 - 3xy^2 - 3y + x = 0$, and the integral is $3\tan(\frac{\arctan(x)}{3})$ (see examples). We present in this paper an algorithm, implemented in the Scratchpad computer algebra system, which generalizes this process and does not require the use of complex logarithms and exponentials.

THE REAL VERSION OF THE STRUCTURE THEOREM

We first need a few definitions in order to formalize the notion of a real elementary function. A *differential field* is a field k with a given map $a \to a'$ from k into k, satisfying $(a+b)' = a' + b'$ and $(ab)' = a'b + ab'$. Such a map is called a *derivation* on k. An element $a \in k$ which satisfies $a' = 0$ is called a *constant*. The constants of k form a subfield of k, denoted by k^{const}. A differential field K is a *differential extension of* k if $k \subseteq K$, and the derivation on K extends the one on k. Let K be a differential extension of k, and $\theta \in K$. We say that θ is *real elementary over* k if either θ is algebraic over k, or there exists $\eta \in k$ such that either

[1] We write $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{C}$ for the fields of rational, real, and complex numbers.

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(i) $\theta' = \frac{\eta'}{\eta}$ (i.e. $\theta = \log(\eta)$),
(ii) $\theta' = \eta'\theta$ (i.e. $\theta = \exp(\eta)$),
(iii) $\theta' = \eta'(1+\theta^2)$ (i.e. $\theta = \tan(\eta)$),
(iv) $\theta' = \frac{\eta'}{1+\eta^2}$ (i.e. $\theta = \arctan(\eta)$).

Note that θ can satisfy one of $(i) - (iv)$ and also be algebraic over k.

A differential extension K of k is a *real elementary extension of* k, if there exist $\theta_1, \ldots, \theta_m \in K$ such that $K = k(\theta_1, \ldots, \theta_m)$ and θ_i is real elementary over $k(\theta_1, \ldots, \theta_{i-1})$ for $i = 1 \ldots m$.

Risch proved the following Theorem which gives a necessary and sufficient condition for a real elementary function over a differential field F of characteristic 0 to be algebraic over F.

THEOREM (RISCH 1979). *Let $K = \mathbf{R}(x, \theta_1, \ldots, \theta_m)$ be a real elementary extension of $\mathbf{R}(x)$, where $K^{const} = \mathbf{R}$, and $x' = 1$. Let $z_1 = e^{y_1}, \ldots, z_p = e^{y_p}$, $y_{p+1} = \log(z_{p+1}), \ldots, y_r = \log(z_r)$, $v_1 = \tan(u_1), \ldots, v_s = \tan(u_s)$, $u_{s+1} = \arctan(v_{s+1}), \ldots, u_t = \arctan(v_t)$ be the quantities among the θ_j's which satisfy $(i) - (iv)$. Let θ be real elementary over K. If θ is algebraic over K, then*

(1) *if $\theta = y = \log(z)$ for $z \in K$, or $\theta = z = e^y$ for $y \in K$, then there exist $c \in \mathbf{R}$, $c_1, \ldots, c_r \in \mathbf{Q}$ such that $y + \sum_{i=1}^{r} c_i y_i = c$,*
(2) *if $\theta = v = \tan(u)$ for $u \in K$, or $\theta = u = \arctan(v)$ for $v \in K$, then there exist $c \in \mathbf{R}$, $c_1, \ldots, c_s \in \mathbf{Q}$ such that $u + \sum_{i=1}^{s} c_i u_i = c$.*

In order to make an algorithm out of this Theorem, we need to compute an algebraic relation for θ once c and the c_i's are known. For part (1), we use Risch's algorithm as in the complex case, while for part (2), we use the following formula:

CLAIM (FLORY 1978). *Let $a_1, \ldots, a_n \in \mathbf{C}$ be such that $\tan(a_i)$ and $\tan(\sum_{i=1}^{n} a_i)$ are defined. Let $S_1, \ldots, S_n$ be the elementary symmetric polynomials in $\tan(a_1)$, $\ldots, \tan(a_n)$, i.e.*

$$\prod_{i=1}^{n}(X - \tan(a_i)) = X^n - S_1 X^{n-1} + \cdots + (-1)^n S_n.$$

Then,

$$\tan(\sum_{i=1}^{n} a_i) = \frac{\sum_{j \text{ odd}} (-1)^{(j-1)/2} S_j}{1 + \sum_{j \text{ even}} (-1)^{j/2} S_j} = \frac{S_1 - S_3 + S_5 + \cdots}{1 - S_2 + S_4 + \cdots}$$

PROOF: By induction on n. For $n = 1$, we have $S_1 = \tan(a_1)$ so the claim is verified. Suppose the claim true for $n \geq 1$, and let

$$\sigma_o = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} (-1)^{(j-1)/2} S_j$$

and

$$\sigma_e = \sum_{\substack{2 \leq j \leq n \\ j \text{ even}}} (-1)^{j/2} S_j$$

where $S_1, \ldots, S_n$ are the elementary symmetric polynomials in $\tan(a_1), \ldots, \tan(a_n)$. Then,

$$\begin{aligned}\tan(\sum_{i=1}^{n+1} a_i) &= \frac{\tan(\sum_{i=1}^{n} a_i) + \tan(a_{n+1})}{1 - \tan(\sum_{i=1}^{n} a_i)\tan(a_{n+1})} \\ &= \frac{\frac{\sigma_o}{1+\sigma_e} + \tan(a_{n+1})}{1 - \frac{\sigma_o}{1+\sigma_e}\tan(a_{n+1})} \\ &= \frac{\sigma_o + \tan(a_{n+1})(1+\sigma_e)}{1 + \sigma_e - \tan(a_{n+1})\sigma_o}.\end{aligned}$$

Let

$$\tau_o = \sum_{\substack{1 \leq j \leq n+1 \\ j \text{ odd}}} (-1)^{(j-1)/2} T_j$$

and

$$\tau_e = \sum_{\substack{2 \leq j \leq n+1 \\ j \text{ even}}} (-1)^{j/2} T_j$$

where $T_1, \ldots, T_{n+1}$ are the elementary symmetric polynomials in $\tan(a_1), \ldots, \tan(a_{n+1})$. From their definition we have $T_1 = \tan(a_{n+1}) + S_1$, $T_{n+1} = S_n \tan(a_{n+1})$, and $T_j = S_j + \tan(a_{n+1}) S_{j-1}$ for $j = 2 \ldots n$. Thus, $\tan(a_{n+1}) + \sigma_o + \tan(a_{n+1})\sigma_e = \tau_o$ and $\sigma_e - \tan(a_{n+1})\sigma_o = \tau_e$, so

$$\tan(\sum_{i=1}^{n+1} a_i) = \frac{\tau_o}{1 + \tau_e}$$

so the claim is verified for $n + 1$.

As a consequence we show how, given $a \in \mathbf{C}$ and $c \in \mathbf{Q}$ such that $\tan(a)$ and $\tan(ca)$ are defined, we can find $P, Q \in \mathbf{Z}[X]$ and $H \in \mathbf{Z}[X, Y]$ such that $H(\tan(a), \alpha) = 0$ and $\tan(ca) = \frac{P(\alpha)}{Q(\alpha)}$.

Write $c = \frac{p}{q}$ where $p, q \in \mathbf{Z}$, and $q > 0$. If $p = 0$, then $\tan(ca) = 0$ so $P = 0$ and $Q = 1$ are sufficient. If $p < 0$, then $\tan(ca) = -\tan(\frac{-p}{q}a)$, so we can assume that $p > 0$. Applying the above claim to $a_i = \frac{a}{q}$ for $i = 1 \ldots q$ we get

$$\tan(a) = \frac{A(\tan(a/q))}{D(\tan(a/q))}$$

where $A, D \in \mathbf{Z}[X]$. Thus, $H(\tan(a), \tan(a/q)) = 0$ where $H(X,Y) = XD(Y) - A(Y) \in \mathbf{Z}[X,Y]$. Applying the claim again to $a_i = \dfrac{a}{q}$ for $i = 1 \ldots p$ we get

$$\tan(ca) = \frac{P(\tan(a/q))}{Q(\tan(a/q))} = \frac{P(\alpha)}{Q(\alpha)}$$

where $P, Q \in \mathbf{Z}[X]$, and $H(\tan(a), \alpha) = 0$. Thus, we can write $\tan(ca)$ as a rational function in a known algebraic function of $\tan(a)$. We note that, since the above $H(X,Y)$ always has degree 1 in X, $\tan(a)$ is an algebraic function of genus 0 over $\tan(a/q)$, so rationalizing substitutions are possible (see examples).

The algorithm

Let $f(x)$ be a given real elementary function in the variable x. We build a tower $F = K(x, \theta_1, \ldots, \theta_n)$ where $F^{const} = K \subset \mathbf{R}$ is a finitely generated extension of $\mathbf{Q}$, the algebraic structure of K over $\mathbf{Q}$ is known, θ_i is real elementary over $k_i = K(x, \theta_1, \ldots, \theta_{i-1})$ for $i = 1 \ldots n$, and $f \in F$. The following algorithm, which determines whether θ_i is algebraic over k_i when the algebraic structure of k_i over $\mathbf{Q}$ is known is applied first to $k_1 = K(x)$, then to k_2, etc... until the algebraic structure of F over $\mathbf{Q}$ is known. As in the complex case, this algorithm will fail if the constants introduced at various stages cause the structure of the constant field over $\mathbf{Q}$ to become unknown. In addition, if a non-real complex constant is introduced, then the above Theorem becomes invalid and we have to revert to its complex version. Such considerations do not appear to be limitations in practice.

The logarithm and exponential cases were given by Risch (1979) and are repeated here for completeness.

Case 1: $\theta_i = y = \log(z)$ for $z \in k_i$.

Let $z_1 = e^{y_1}, \ldots, z_p = e^{y_p}$, $y_{p+1} = \log(z_{p+1}), \ldots, y_r = \log(z_r)$ be the exponentials and logarithms in k_i. By Risch's theorem, if θ_i is algebraic over k_i, then there exist $c \in \mathbf{R}$, $c_1, \ldots, c_r \in \mathbf{Q}$ such that

$$y + \sum_{i=1}^{r} c_i y_i = c.$$

Applying $' = \dfrac{d}{dx}$ we get

$$\frac{z'}{z} + \sum_{i=1}^{r} c_i \frac{z_i'}{z_i} = 0. \tag{1}$$

Since $z_1, \ldots, z_r, z \in k_i$, and a vector space basis for k_i over $\mathbf{Q}$ is known, (1) reduces to a system of linear equation Δ_1 for $c_1, \ldots, c_r$ over $\mathbf{Q}$. If Δ_1 has no solution, then θ_i is transcendental over k_i, otherwise, we get

$$e^c = z \prod_{i=1}^{r} z_i^{c_i}$$

which is a constant, and

$$\theta_i = c - \sum_{i=1}^{r} c_i y_i \in K(c)(\theta_1, \ldots, \theta_{i-1}).$$

Case 2: $\theta_i = z = e^y$ for $y \in k_i$.

Let $z_1 = e^{y_1}, \ldots, z_p = e^{y_p}$, $y_{p+1} = \log(z_{p+1}), \ldots, y_r = \log(z_r)$ be the exponentials and logarithms in k_i. By Risch's theorem, if θ_i is algebraic over k_i, then there exist $c \in \mathbf{R}$, $c_1, \ldots, c_r \in \mathbf{Q}$ such that

$$y + \sum_{i=1}^{r} c_i y_i = c.$$

Applying $' = \dfrac{d}{dx}$ we get

$$y' + \sum_{i=1}^{r} c_i y_i' = 0. \tag{2}$$

Since $y_1, \ldots, y_r, y \in k_i$, and a vector space basis for k_i over $\mathbf{Q}$ is known, (2) reduces to a system of linear equation Δ_2 for $c_1, \ldots, c_r$ over $\mathbf{Q}$. If Δ_2 has no solution, then θ_i is transcendental over k_i, otherwise, we get c from the above equation, and

$$\theta_i = e^{c - \sum_{i=1}^{r} c_i y_i} = e^c \prod_{i=1}^{r} z_i^{-c_i} \in \overline{K(e^c)(\theta_1, \ldots, \theta_{i-1})}.$$

Case 3: $\theta_i = v = \tan(u)$ for $u \in k_i$.

Let $v_1 = \tan(u_1), \ldots, v_s = \tan(u_s)$, $u_{s+1} = \arctan(v_{s+1}), \ldots, u_t = \arctan(v_t)$ be the tangents and arc tangents in k_i. By Risch's theorem, if θ_i is algebraic over k_i, then there exist $c \in \mathbf{R}$, $c_1, \ldots, c_t \in \mathbf{Q}$ such that

$$u + \sum_{i=1}^{t} c_i u_i = c.$$

Applying $' = \dfrac{d}{dx}$ we get

$$u' + \sum_{i=1}^{t} c_i u_i' = 0. \tag{3}$$

Since $u_1, \ldots, u_t, u \in k_i$, and a vector space basis for k_i over $\mathbf{Q}$ is known, (3) reduces to a system of linear equation Δ_3 for $c_1, \ldots, c_t$ over $\mathbf{Q}$. If Δ_3 has no solution, then θ_i is transcendental over k_i, otherwise, we get c from the above equation, and

$$\begin{aligned}\theta_i &= \tan\left(c - \sum_{i=1}^{t} c_i u_i\right) \\ &= \frac{P(\tan(c), \tan(-c_1 u_1), \ldots, \tan(-c_t u_t))}{Q(\tan(c), \tan(-c_1 u_1), \ldots, \tan(-c_t u_t))}\end{aligned}$$

where P, Q are polynomials given by the claim. Since $c_i \in \mathbf{Q}$ we can express $\tan(-c_i u_i)$ as a rational function in α_i, where α_i satisfies an algebraic relation $H_i(v_i, \alpha_i) = 0$, so we get a formula for $\theta_i \in \overline{K(\tan(c))(\theta_1, \ldots, \theta_{i-1})}$.

Case 4: $\theta_i = u = \arctan(v)$ **for** $v \in k_i$.

Let $v_1 = \tan(u_1), \ldots, v_s = \tan(u_s)$, $u_{s+1} = \arctan(v_{s+1}), \ldots, u_t = \arctan(v_t)$ be the tangents and arc tangents in k_i. By Risch's theorem, if θ_i is algebraic over k_i, then there exist $c \in \mathbf{R}$, $c_1, \ldots, c_t \in \mathbf{Q}$ such that

$$u + \sum_{i=1}^{t} c_i u_i = c.$$

Applying $' = \dfrac{d}{dx}$ we get

$$\frac{v'}{1+v^2} + \sum_{i=1}^{r} c_i {u_i}' = 0. \tag{4}$$

Since $u_1, \ldots, u_t, v \in k_i$, and a vector space basis for k_i over $\mathbf{Q}$ is known, (4) reduces to a system of linear equation Δ_4 for $c_1, \ldots, c_t$ over $\mathbf{Q}$. If Δ_4 has no solution, then θ_i is transcendental over k_i, otherwise, we get

$$\begin{aligned}\tan(c) &= \tan(u + \sum_{i=1}^{t} c_i u_i) \\ &= \frac{P(v, \tan(c_1 u_1), \ldots, \tan(c_t u_t))}{Q(v, \tan(c_1 u_1), \ldots, \tan(c_t u_t))}\end{aligned}$$

which is a constant, and where P, Q are polynomials given by the claim. Using the expression for $\tan(c_i u_i)$ in terms of α_i as above, we get a constant expression for $\tan(c)$, and

$$\theta_i = c - \sum_{i=1}^{t} c_i u_i \in K(c)(\theta_1, \ldots, \theta_{i-1}).$$

EXAMPLES

1. Consider

$$f = \frac{\tan(\dfrac{\arctan(x)}{3})^2 + 1}{x^2+1} \in \mathbf{Q}(x, u_1, v_2)$$

where $u_1 = \arctan(x)$ and $v_2 = \tan(\arctan(x)/3)$. Let $k_1 = \mathbf{Q}(x)$, $\theta_1 = u_1 = \arctan(v_1)$ where $v_1 = x \in k_1$. Equation (4) is $\dfrac{1}{1+x^2} = 0$ which has no solution, so $\arctan(x)$ is transcendental over k_1. Let $k_2 = \mathbf{Q}(x, u_1)$ and $\theta_2 = v_2$. We have $u_2 = \arctan(x)/3$, so equation (3) is

$$\frac{1}{3+3x^2} + \frac{c_1}{1+x^2} = 0$$

which has $c_1 = -\dfrac{1}{3}$ for solution, so we get $c = u_2 - \dfrac{1}{3}u_1 = 0$, so $v_2 = \tan(u_2) = \tan(u_1/3)$. From the claim we get

$$x = \tan(u_1) = \frac{\tan(u_1/3)^3 - 3\tan(u_1/3)}{3\tan(u_1/3)^2 - 1}$$

so $y = v_2$ is an algebraic function satisfying $y^3 - 3xy^2 - 3y + x = 0$ and

$$f = \frac{y^2+1}{x^2+1} \in \mathbf{Q}(x, y).$$

Substituting $x = \dfrac{y^3 - 3y}{3y^2 - 1}$, $dx = \dfrac{3y^4 + 6y^2 + 3}{9y^4 - 6y^2 + 1}dy$, we get

$$f = \frac{9y^4 - 6y^2 + 1}{y^4 + 2y^2 + 1} \in \mathbf{Q}(y)$$

so

$$\int f dx = \int 3dy = 3y = 3\tan(\frac{\arctan(x)}{3}).$$

2. Consider

$$f = \tan(x + 2 - \frac{3}{4}\arctan(x)) - \tan(x) \in \mathbf{Q}(x, u_1, v_2, v_3)$$

where $u_1 = \arctan(x)$, $v_2 = \tan(x)$, and $v_3 = \tan(x + 2 - 3\arctan(x)/4)$. Let $k_1 = \mathbf{Q}(x)$, $\theta_1 = u_1 = \arctan(v_1)$ where $v_1 = x \in k_1$. Equation (4) is $\dfrac{1}{1+x^2} = 0$ which has no solution, so $\arctan(x)$ is transcendental over k_1. Let $k_2 = \mathbf{Q}(x, u_1)$ and $\theta_2 = v_2$. We have $u_2 = x$, so equation (3) is

$$1 + \frac{c_1}{1+x^2} = 0$$

which has no solution, so θ_2 is transcendental over k_2. Let $k_3 = \mathbf{Q}(x, u_1, v_2)$ and $\theta_3 = v_3$. We have $u_2 = x$ and $u_3 = 2 + x - 3\arctan(x)/4$ so equation (3) is

$$1 - \frac{3}{4+4x^2} + \frac{c_1}{1+x^2} + c_2 = 0$$

which has $(c_1, c_2) = (\dfrac{3}{4}, -1)$ for solution, so we get $c = u_3 + 3u_1/4 - u_2 = 2$ and

$$\tan(u_3) = \tan(2 - \frac{3}{4}u_1 + u_2) = \frac{A}{B}$$

where

$$\begin{aligned}A = \tan(2) - \tan(3u_1/4) + \tan(u_2) \\ + \tan(2)\tan(3u_1/4)\tan(u_2)\end{aligned}$$

and

$$\begin{aligned}B = 1 - \tan(2)\tan(u_2) + \tan(2)\tan(3u_1/4) \\ + \tan(u_2)\tan(3u_1/4).\end{aligned}$$

From the claim we get

$$x = \tan(u_1) = \frac{4\tan(u_1/4) - 4\tan(u_1/4)^3}{1 - 6\tan(u_1/4)^2 + \tan(u_1/4)^4}$$

so $y = \tan(u_1/4)$ is an algebraic function satisfying

$$xy^4 + 4y^3 - 6xy^2 - 4y + x = 0$$

and, as in the previous example,

$$\tan(3u_1/4) = \frac{y^3 - 3y}{3y^2 - 1}$$

and replacing in the above expression for $\tan(u_3)$, we get an expression for f in $\mathbf{Q}(\tan(2))(x, y, \tan(x))$. Substituting $x = \dfrac{4y - 4y^3}{1 - 6y^2 + y^4}$, we get an expression for f in $\mathbf{Q}(\tan(2))(y, \tan(\dfrac{4y - 4y^3}{1 - 6y^2 + y^4}))$, which is purely transcendental.

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